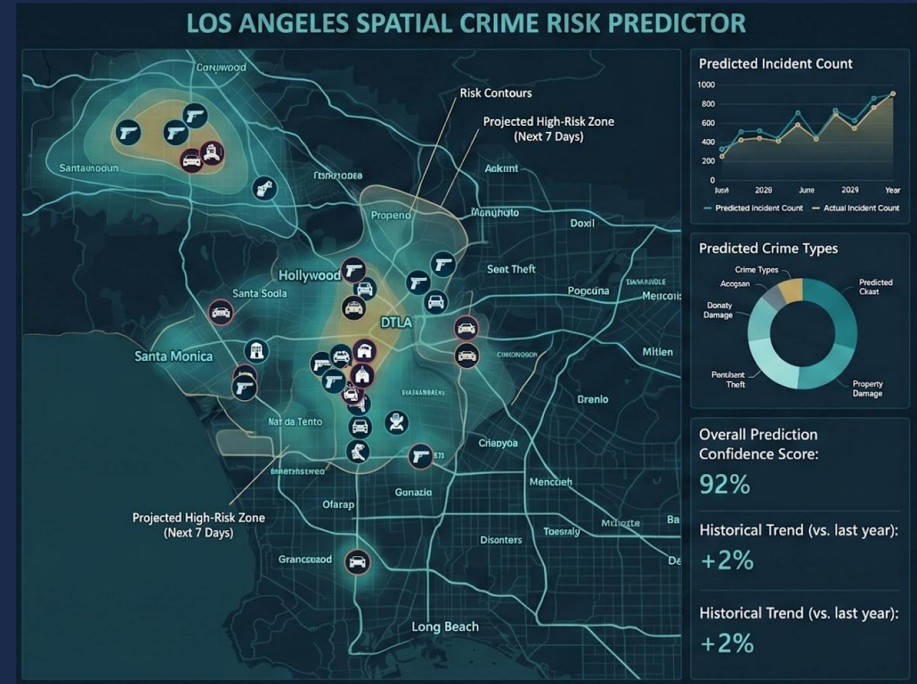


SafeCity Analytics

Predictive Crime Analysis of Los Angeles

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EAS 587: Data-Intensive Computing · Spring 2026 · University at Buffalo



What We Built

Phase 1

Data & EDA

62K LAPD crime records · 10
cleaning operations · 10 EDA
analyses

Phase 2

ML Models

6 algorithms · Crime category,
severity, weapon risk, hotspots
· MCP server deployed

Phase 3

Spark & Databricks

Databricks Delta Lake · NIBRS
integration · Bronze-Silver-Gold
pipeline

Phase 4

Presentation

Visualization · Complete
pipeline synthesis

Problem Statement

Most crime dashboards describe the past. We predict the future.

Crime Category Prediction

Given time, location, and context — what type of crime is this? Supports dispatch pre-fill and officer prep.

Weapon-Involvement Risk

Does this incident carry elevated weapon risk? Calibrated probability score for officer safety decisions.

Geographic Hotspot Discovery

Which clusters account for highest density? Supports patrol scheduling and resource allocation.

Data Collection & Cleaning

62,117

LAPD Crime Records
(2024)

28

Raw Columns

10

Cleaning Operations

46

Final Features

Key Cleaning Operations:

1. Date parsing + Hour / Month / DayOfWeek extraction
2. Victim age=0 → NaN (unknown, not infant)
3. 140+ crime codes → 12 semantic categories
4. Crm Cd 2/3/4 removed (98–100% missing)
5. Reporting delay computed (days to report)

Key EDA Findings:

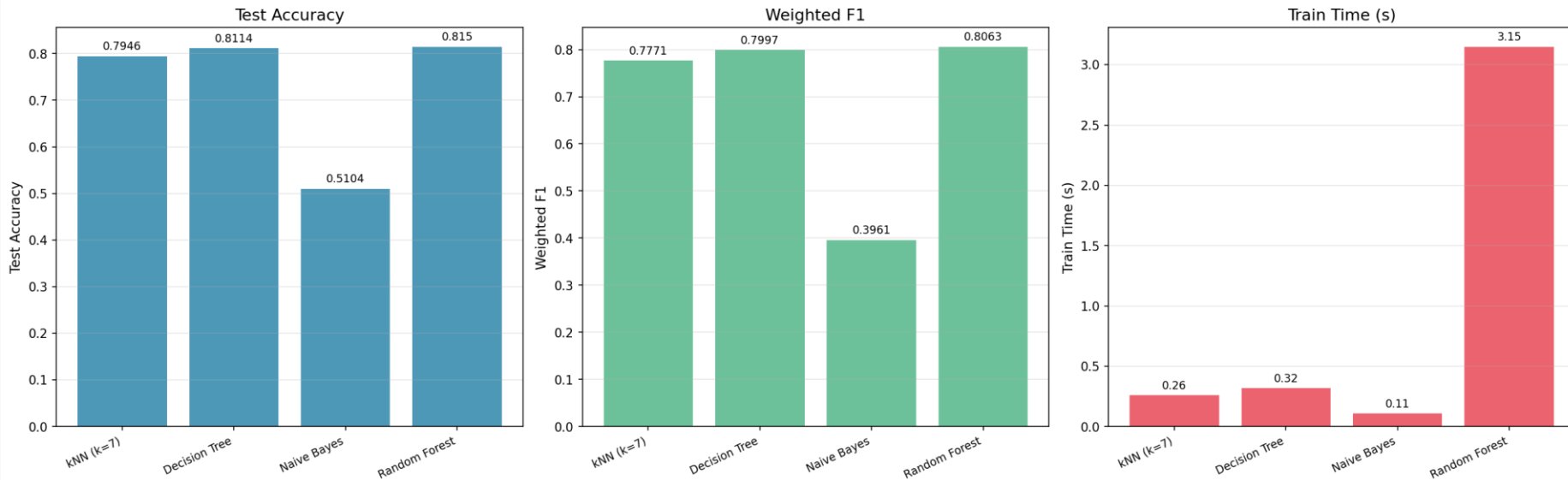
- Peak crime: 6 PM · Highest day: Friday
- Vehicle Crime = 46.2% of all incidents
- Only 5.9% of crimes involve a weapon
- Central LA: 6,024 crimes (highest area)
- 75% of crimes reported within 3 days

Six ML Algorithms

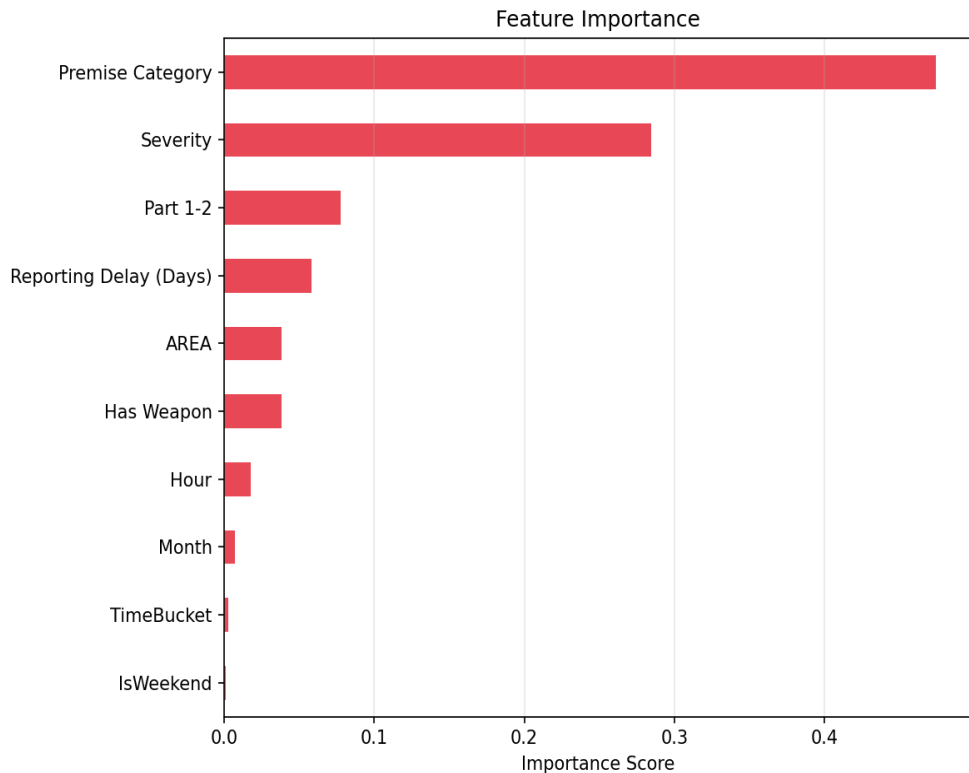
kNN	In-class	Crime Category (13-class)	k=7, StandardScaler
Decision Tree	In-class	Crime Category (13-class)	GridSearchCV tuning
k-Means	In-class	Geographic Hotspots	k=2, silhouette guided
Naive Bayes	In-class	Crime Category (13-class)	ComplementNB selected
Random Forest	Outside class	Crime Category + MCP server	RandomizedSearchCV · balanced weights
Logistic Reg.	Outside class	Weapon Involvement (binary)	Calibrated probability scoring

Model Performance Summary

Algorithm Comparison — Crime Category Classification



Decision Tree: Feature Importances



Findings

Premise Category drives the tree (~45%)

Where a crime occurs is the strongest single predictor of its type — land use and patrol district structure dominate.

Severity matters (~28%)

Crime classification level significantly influences tree splits.

Part 1-2 & Reporting Delay carry modest weight (~6-7%)

Supplementary administrative features provide limited additional signal.

Minority classes struggle

Homicide and Arson show $F1 \approx 0$. With <20 test samples, even a perfect model can't demonstrate recall.

Geographic Crime Hotspots

k-Means Clustering ($k=2$, Silhouette = 0.546)

Cluster 0 — West / Coastal

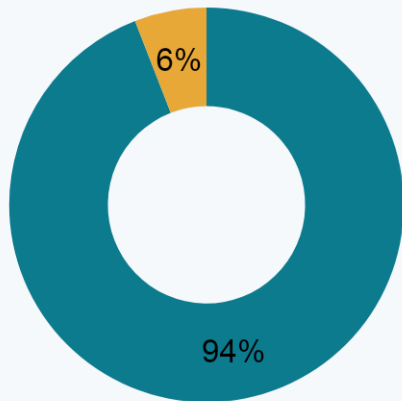
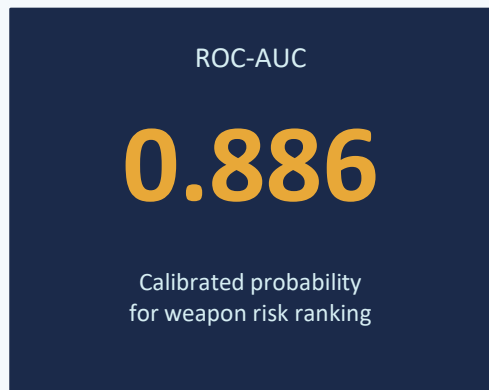
- › Districts: West LA, Pacific, Hollywood, Wilshire
- › Top crime: Vehicle Crime (44%)
- › Notable: High Theft rate in commercial zones
- › Reporting delay: avg 2.1 days

Cluster 1 — Central / Inland

- › Districts: Central, Newton, 77th St, Southwest
- › Top crime: Vehicle Crime (48%) — slightly higher
- › Notable: Higher Assault share vs Cluster 0
- › Central LA alone: 6,024 incidents (most)

DBSCAN was tried — labeled 60%+ as noise. LA crime is spatially diffuse; k-Means was retained.

Logistic Regression: Weapon-Involvement Risk



■ No Weapon (94.1%) ■ Weapon (5.9%)

Why This Matters

Base rate: 5.9%

Weapon crimes are rare — a hard target. Precision on the weapon class is 0.26.

Probabilities > labels

Officers need a risk score, not a binary guess. Threshold can be tuned by district.

ROC-AUC = 0.886

Model ranks high-risk incidents well — the top 10% of scores capture most weapon cases.

MCP: Natural Language → Crime Prediction

How It Works

1

Register

```
claude mcp add -s project safecity-crime-predictor -- python3  
src/mcp/server.py
```

2

Model loads

Random Forest (pkl) loaded; auto-retrains from CSV if missing

3

User asks Claude

"What crime is likely at 11 PM on a public street in area 13?"

4

Tool called

`predict_crime_category` → returns category + top-3 probability breakdown

Exposed Tools

`predict_crime_category`

Input: area, hour, month, weekend, weapon, premise, time bucket, severity, part, reporting delay

Output: category label + top-3 probability breakdown

`list_crime_categories`

Returns all 13 supported crime category labels

`server_health`

Returns model status, feature count, and model path

Phase 3: Spark + Delta Lake Architecture



MCP Inference Server (Phase 2) · `src/mcp/server.py` · `predict_crime_category` · `list_crime_categories` · `server_health`

PHASE 3 — NIBRS Integration & MLlib Results

8 Databricks Notebooks | Cross-Source Join on CaseNo | 3 NIBRS Insights | Spark MLlib Pipeline Saved

8-Notebook Databricks Pipeline

LAPD Pipeline (Notebooks 01-04)

- 01 Bronze — raw CSV to bronze_mydata Delta table
- 02 Silver — clean, transform, derive features
- 03 Gold — 6 business aggregate Delta tables
- 04 MLlib — Decision Tree + Naive Bayes (Spark)

NIBRS Pipeline (Notebooks 05-08)

- 05 Bronze — NIBRS Victims (232K) + Offenses (250K)
- 06 Silver — left-join LAPD + NIBRS via CaseNo
- 07 Insights — 3 cross-source statistical analyses
- 08 MLlib LR — weapon prediction on merged data

Insight 1: NIBRS Confirms LA Vehicle Crime Spike

LAPD vehicle crime rate (46.2%) exceeds the national NIBRS baseline. Cross-joining both sources on CaseNo shows the pattern holds independently in federal data — not an artifact of LAPD classification alone.

Insight 2: Temporal Patterns Agree Across Sources

NIBRS hourly distribution (250K offenses) mirrors LAPD peak at 6 PM. Day-of-week patterns align across both agencies. Two independent sources, same signal — validates Phase 2 temporal features for ML.

Insight 3: Spark MLlib on Merged NIBRS+LAPD Data

Notebook 08 trains a Spark MLlib Logistic Regression on silver_lapd_crimes — adding NIBRS-specific features: DomesticViolence, GangRelated, HomelessVictim. Saved as full VectorAssembler + StandardScaler + LR pipeline to Databricks volumes for reproducible reuse.

Gold Tables: area summary, monthly trends, hourly patterns (weekday/weekend), victim profiles, hotspot features, ML features

Scale: 232K NIBRS victims + 250K NIBRS offenses joined to 62K LAPD records via CaseNo left join

Key Insights Across the Pipeline

01 Crime is structurally predictable

82% category accuracy from area, hour, and premise type alone. Most variance is tied to land use and time — not randomness.

02 Weapon risk ranking works; weapon detection does not

ROC-AUC 0.886 means the model ranks incidents by risk well. But with only 5.9% positive rate, binary precision is 0.26. Use thresholds, not hard labels.

03 LA crime geography is east-west, not cluster-dense

k=2 beats finer granularity. The coastal vs. inland split is real and stable. DBSCAN failed because crime is diffuse — not point-clustered.

04 Shared preprocessing prevents silent bugs

A single preprocess.py module across all six models eliminated per-script encoding drift and made the Spark pipeline reproducible.

Limitations & Future Work

Current Limitations

Single year (2024) — no seasonal or trend analysis possible

48.5% unknown victim demographics limits demographic modeling

13-category label scheme loses fine-grained crime distinctions

Next Steps

→ Multi-year data

2020–2024 LAPD records would unlock seasonal trends and year-over-year baselines.

→ Hierarchical crime taxonomy

Replace flat 13-class labels with a hierarchy — enables shared structure between related crimes.

→ Spark Structured Streaming

Connect MCP server to live LAPD API feed for near-real-time dispatch inference.

→ Fairness audit

Evaluate model predictions for disparate impact across victim descent groups.

SafeCity Analytics

End-to-End Crime Prediction Pipeline

81.5%

Crime Category
Accuracy (RF)

0.886

Weapon Risk
ROC-AUC (LR)

0.546

Cluster
Silhouette

- Complete pipeline: raw CSV → EDA (Ph.1) → 6 ML models + MCP server (Ph.2) → Spark/Databricks (Ph.3)
- 6 ML models covering classification, clustering, and risk scoring
- Random Forest deployed as MCP server callable from Claude Code
- NIBRS cross-source validation confirms LA vehicle crime above national baseline

Any Questions ?